

Engineering Change Notice(ECN)

ECN-D-2601-01(Side Ion Bar Mounting Hole Location Change

(ECP AP Type Down Buffer/Aligner))

FA연구소

작성	검토	승인
곽수민	주성진	이태경
26/01/05	26/01/05	26/01/05

NINEBELL

1. Engineering Change Request

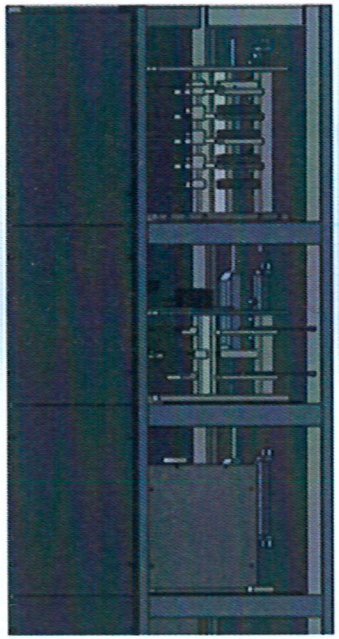
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설계 변경 요청서(ECR)
(Engineering Change Request)

Customer	ACM		
ECR No.	ECR-S-2512-10		

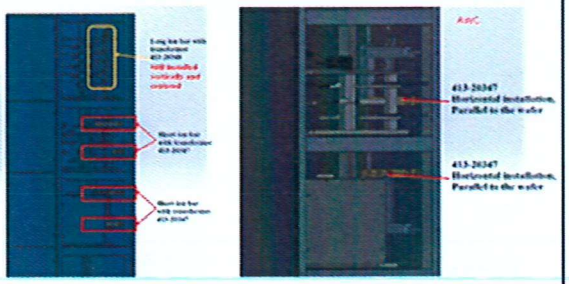
Requesting department	Name / Department	CHENGYANG JIANG	변경 사유	☑ Customer Request ☐ Cost Reduction ☐ Quality improve ☐ Work Efficiency ☐ Discontinued ☐ 기타	
	Date	2025.11.30			
Request project	Category	Project Type	변경 Level (Change Level)	☑ Level 1	Design structure changes are required
	☑ MP ☐ ES	ECP AP(ECN 전달 시, 가장 빨리 적용 가능한 AP 타입 장비 명시 필요)		☐ Level 2	Simple part and work method change

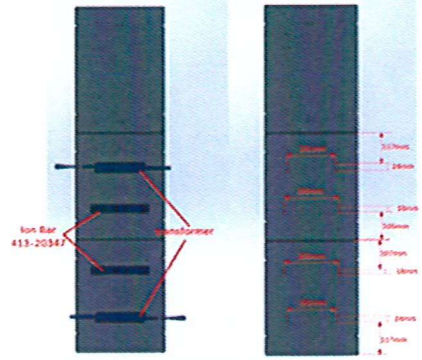
Before(Problems and Change Requests)



The ESD Tests for Buffer and Aligner Areas were Failed at FAB, therfore we request to adjust the installation position of the Ion Bars to pass the ESD Tests.

After(Expected Effect)

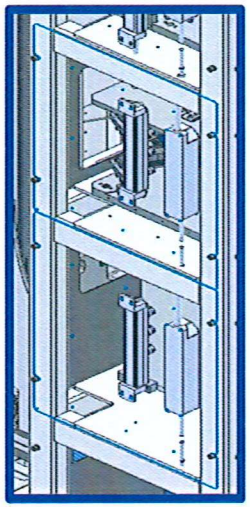
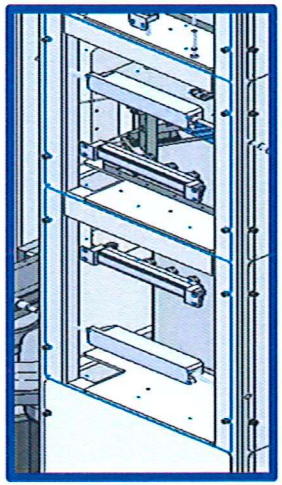




Adjusting the installation position of the Ion Bars. The ion bar(413-20348) in the upper buffer area remains vertically centered. The ion bars(413-20347) in the rest of two areas are adjusted to be horizontally installed.
The ion bar outlet of the middle buffer area is level with the wafer.
Each ion bar comes with a transformer which inculded in the ion bar. The Distance between transformer and ion bar is 0.8 meters. Each ion bar needs a controller (413-20354). The actual distance between transformer and controller might be too long for orignial cable length(0.3 meters), Therfore there are extend cable provided by Keyence(413-20378). The extend cable (3 meters for one unit) can not be used more than 9 metes for each set of ion bar and controller.

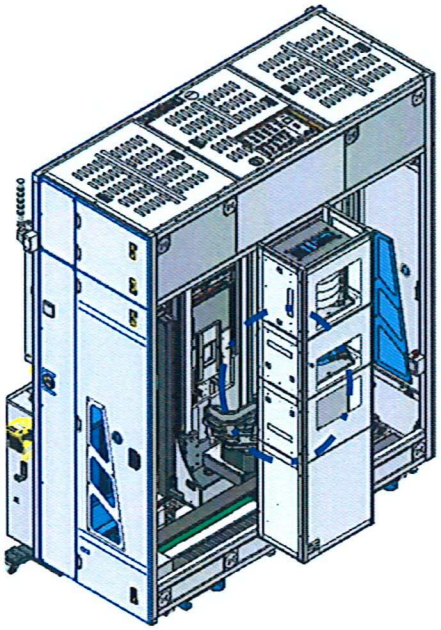
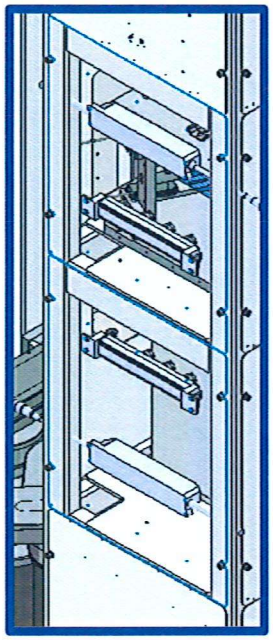
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	ECP AP-L TYPE	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) □ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) □ 기타 (ETC)	ECR NO.	ECR-S-2512-10
		ECN NO.	ECN-D-2601-01
변경 상세 사유 Detailed reason for change	AP TYPE의 EFEM에 전체 적용 ACM에서 현장에서 설치하는 ION BAR를 세로에서 가로 배치로 변경. Planned to be fully applied to AP TYPE EFEM. ACM changed the ION BAR installation on site from vertical to horizontal.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
			
상세내용 Detail	• EEAP-F70P49-0, BUFFER COVER (7) • EEAP-F70P50-0, BUFFER COVER (6)	• EEAP-F70P56-0, BUFFER COVER (9) • EEAP-F70P57-0, BUFFER COVER (10) • Applied in ECP248	

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2. Engineering Change Notice

전체 이미지 Full Image	**VIEW : AP4-L TYPE**   ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approval	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026.01.06 2. Customer : / /		나인벨 승인 Ninebell approval	QC 2026.01.06 PASS	고객사 승인 Customer approval	ACM confirm 1/12/26
적용 일자 Effective date	After approval		최초 적용 장비 Initially Applied Equipment	ECP248(AP4-L-GW)		